

LSTM Variants for Chaotic Dynamical Systems: An Empirical Study on the Lorenz Attractor
Ruslan Gokhman - Yeshiva University - rgokhman@mail.yu.edu

Abstract

Forecasting chaotic dynamical systems such as the Lorenz attractor is notoriously difficult: small numerical errors are amplified exponentially over long autoregressive rollouts. We study seven recurrent and convolutional architectures for the AI-DEEDS 2026 Chaotic Systems Challenge: a vanilla LSTM, an LSTM with additive attention, a Bidirectional LSTM (BiLSTM), a BiLSTM trained with the Huber loss, a Temporal Convolutional Network (TCN), a CNN front-end followed by an LSTM, and a CNN front-end followed by a BiLSTM. All models share the same pre-processing, sequence length, and rollout procedure, isolating the contribution of each design choice. The challenge scores predictions on a 0-100 scale where higher is better. We obtain leaderboard scores between 45.72 and 58.81, with the BiLSTM trained with Huber loss being the strongest configuration.

Keywords: Chaotic systems, Lorenz attractor, LSTM, BiLSTM, Huber loss, time-series forecasting

1 Introduction

The Lorenz 63 system is a canonical example of deterministic chaos: a three-dimensional ODE whose trajectories never repeat and depend sensitively on initial conditions. Tiny errors in the predicted state are amplified exponentially by the positive Lyapunov exponent.

2 Related Work

Common Task Frameworks for scientific ML. The AI-DEEDS competition is part of a growing family of Common Task Frameworks (CTFs) designed to replace ad hoc comparisons.

Physics-informed and hybrid approaches. SINDy recovers sparse polynomial governing equations directly from data and produces interpretable models that generalize well.

Robust losses for time-series. The Huber loss has a long history in robust statistics and has been applied to neural network training to reduce sensitivity to outliers.

3 Problem Setting

The competition provides several training trajectories. The task is partitioned into nine evaluation pairs, each specifying training trajectory, initial window, and forecast steps (1,000 or 10,000).

Two features make the task especially challenging:

- Long horizons: Pairs 2 and 4 require 10,000 autoregressive steps.
- Heterogeneous regimes: Pairs 6 and 7 use different parameter regimes.

4 Methods

Pre-processing and shared pipeline. For each pair the relevant training trajectory is concatenated and standardized. We build supervised pairs with sliding window of length $L = 50$.

(A) Vanilla LSTM. A 2-layer unidirectional LSTM with hidden size 128.

(B) LSTM + Attention. Augments (A) with a Bahdanau-style additive attention pooling over time.

(C) Bidirectional LSTM (BiLSTM). Replaces the LSTM in (A) with a bidirectional one.

(D) BiLSTM + Huber loss. Same architecture as (C), but trained with the Huber loss with $\delta = 1.0$.

- (E) Temporal Convolutional Network (TCN). A 4-layer dilated causal CNN with kernel size 3.
- (F) CNN + LSTM. A two-layer 1D CNN front-end followed by a 2-layer unidirectional LSTM.
- (G) CNN + BiLSTM. Identical to (F) but with a bidirectional LSTM.

5 Results

Table 1: AI-DEEDS 2026 leaderboard scores (higher is better; 100 is perfect):

- (A) Vanilla LSTM: Score 56.81, Mean Loss 0.0550
- (B) LSTM + Attention: Score 45.72, Mean Loss 0.1307
- (C) BiLSTM (MSE): Score 58.16, Mean Loss 0.0289
- (D) BiLSTM + Huber loss: Score 58.81, Mean Loss 0.0149 (Best)
- (E) TCN + Huber: Score 46.55, Mean Loss 0.0272
- (F) CNN + LSTM + Huber: Score 54.19, Mean Loss 0.0405
- (G) CNN + BiLSTM + Huber: Score 53.52, Mean Loss 0.0280

The BiLSTM with Huber loss is the strongest configuration on every metric.

Per-pair RMSE on pair 1 (Lorenz 63 baseline). Table 2 drills into pair 1.

The unidirectional LSTM and the BiLSTM + Huber both achieve perfect single-step prediction.

6 Discussion

Why bidirectional context helps. At first glance, BiLSTMs seem ill-suited to autoregressive forecasting because at inference time there is no future beyond the input window. Within each window, however, the backward pass enriches the representation, helping identify the local dynamical mode.

Why Huber outperforms MSE. Chaotic systems produce occasional large prediction errors. MSE squares such residuals and lets them dominate the gradient. Huber loss caps this influence. Why attention degraded performance. Three factors contribute: short window ($L=50$), attention head/MLP increase overfitting capacity, and Lorenz is near-Markovian (most recent is best). Why the CNN front-ends did not help. The CNN parameters increase capacity to overfit and may smooth away the sharp lobe transitions.

7 Conclusion and Future Work

We compared seven LSTM-family forecasters on a chaotic time series benchmark. Bidirectional context combined with Huber loss gives the best leaderboard score (58.81). Standard improvements such as attention and CNN front-ends are not universally beneficial.

References

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